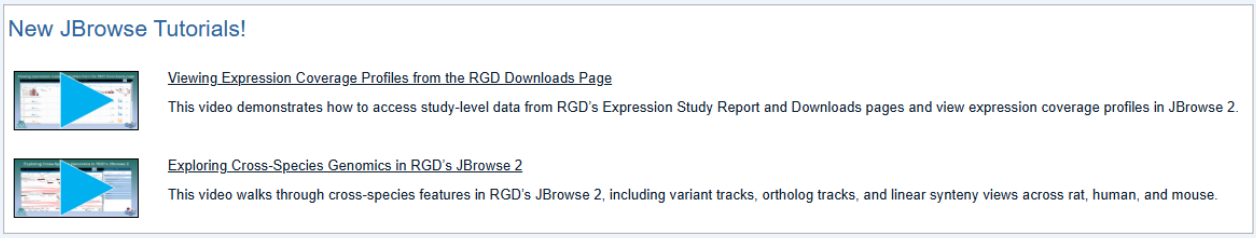
23rd Annual Meeting of the Complex Trait Community

Workshop: Comparative Genomics in JBrowse 2 at the Rat Genome Database (RGD)

November 11, 2026

RGD website: rgd.mcw.edu

RGD Video Tutorials: https://rgd.mcw.edu/wg/home/rgd_rat_community_videos/rgd-tutorials/



JBrowse website: https://jbrowse.org/jb2/

Paper resource: PMID: 37069644

Outline:

- Access JBrowse 2 through RGD
- Navigation
- Variant Tracks
- Ortholog Tracks
- Linear Genome Views
- Human Variant Track
- Synteny Track
- Linear Synteny View
- Download and Open Expression Tracks

> *Note: there was a recent JBrowse 2 update, so your view and what is shown in the workshop may slightly differ, but the overall process and content will be the same.*

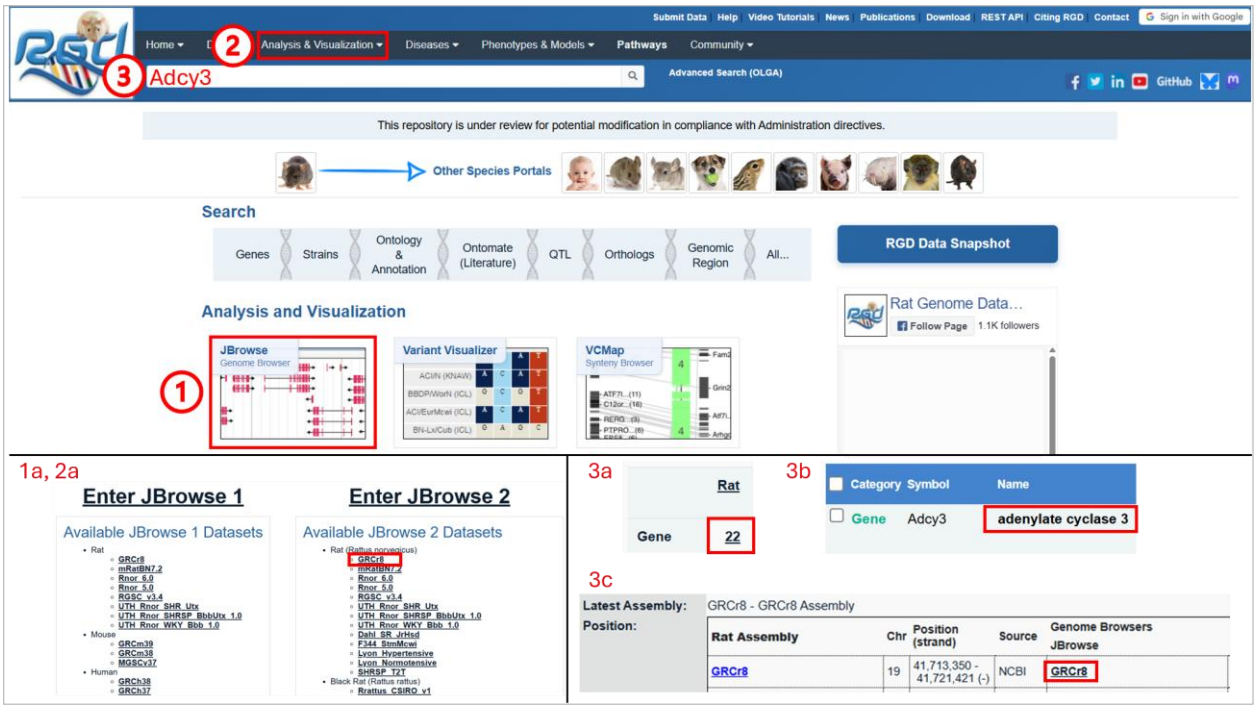
Access JBrowse 2 through RGD (rgd.mcw.edu)

There are multiple ways to access JBrowse from RGD. Here are a few:

1) Homepage → "Analysis and Visualization" section → JBrowse button
2) Homepage → top menu "Analysis & Visualization" tab → "JBrowse (Genome Browser)"
   a) These options will bring you to a new page with options to enter JBrowse 1 or JBrowse 2 by selecting the species assembly of interest. This will open a default view in JBrowse.

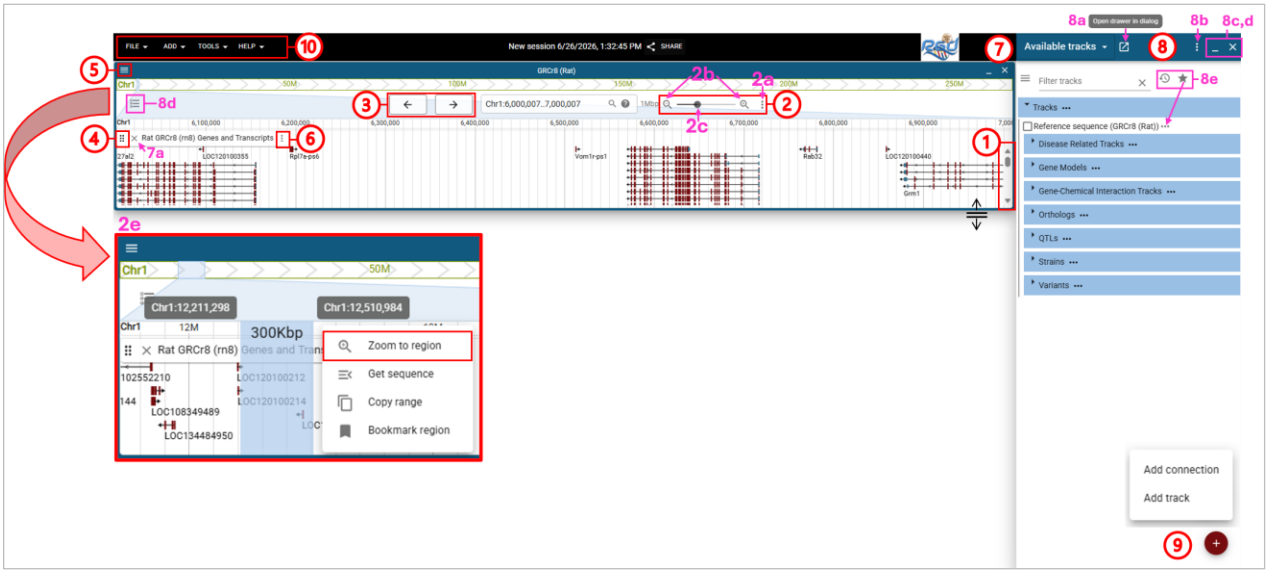
> *Note: **GRCr8 (rn8)** is the most recent assembly for rat, and there are new rat genome annotations now available from NCBI, including annotations for GRCr8 (strain: BN/NHsdMcwi, RRID:RGD_61498) and SHRSP_T2T (strain: SHRSP/BbbUtx, RRID:RGD_8142383).*

3) Homepage → search for a gene (e.g., *Adcy3*)
   a) Select linked number in the gene row for the species of interest (e.g., rat)
   b) Select the gene from the list (e.g., adenylate cyclase 3)
   c) Gene page → "Position" section → "Genome Browsers" → "JBrowse" → select the linked assembly (e.g., GRCr8). This will automatically open to the position of the *Adcy3* gene in JBrowse 2



> *Note: Updates are coming to the RGD Report pages (e.g., Gene page) soon. The layout and content will remain the same, but there will be a new look and feel!*

Navigation



https://rgd.mcw.edu/jbrowse2/?session=share-50A4mlXTRb&password=JxBRU

Adjust track/window size: click and drag when this icon appears ⇳

1) **Scroll** in track up/down with mouse wheel or by clicking the arrows/dragging the bar in the vertical scroll bar
2) **Zoom options:**
   a) 3 vertical dots → zoom in/out options (e.g., Zoom in 100x)
   b) +/- magnifying glass icons
   c) Sliding bar
   d) Ctrl + mouse scroll wheel in a track (not shown)
   e) Click and drag in the detail position bar to select the region → "Zoom to region"
3) **Pan options:**
   a) In a track, click and drag left/right
   b) Left/right arrow buttons scroll left/right

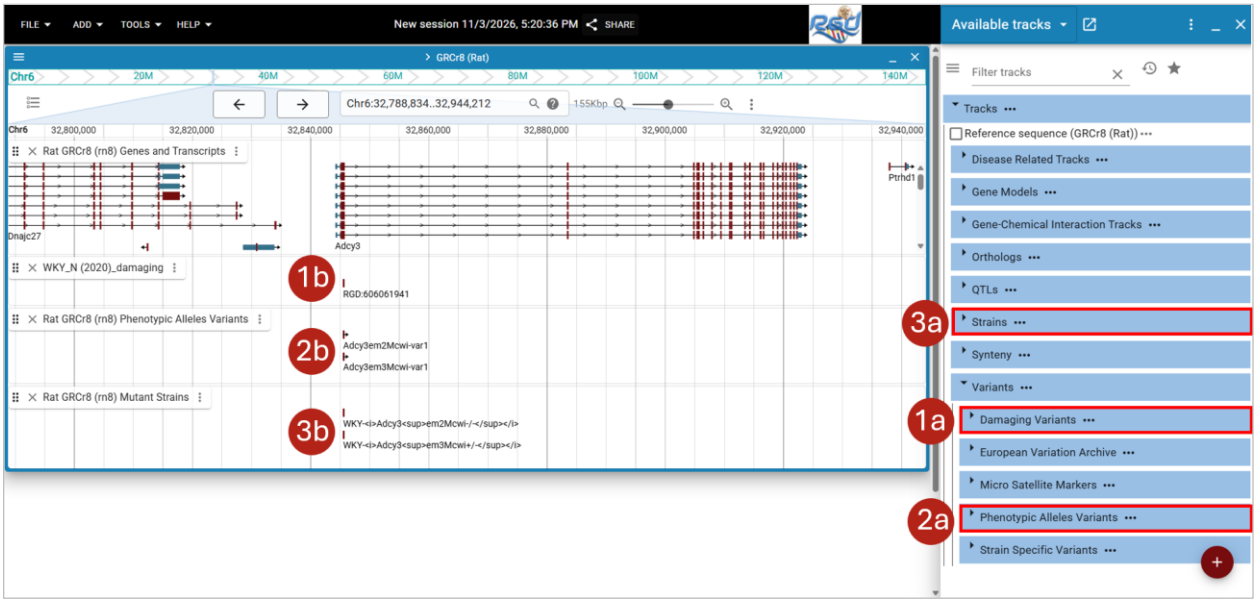
Bonus navigation tips:

4) Re-order tracks: click and drag up/down on the drag handle on the track label (six vertical dots)
5) Re-ordering views: unlike tracks, views cannot be reordered by drag-and-drop. Instead, use the view menu (hamburger icon) and select "Move up"/"Move down"
6) Track menu: About track info, track actions, display settings, and more
7) Each window can be minimized or X'ed out
   a) You can also exit out of individual tracks within the window (see X next to the 4 callout in the track name)

8) Available tracks:
   a) Next to the “Available tracks” title is an icon in which you can open this menu as a dialog box
   b) The vertical 3 dots (right of the 8 label) allow you to toggle the “Available tracks” menu to the left or right of the screen/track window
   c) On the far right of the header, you can minimize the track menu. To re-open, there will be a button on the bottom right of the screen to re-open the track menu
   d) Use the X to completely close the available tracks menu. To re-open, select the hamburger menu (3 horizontal lines beginning with a circle)
   e) Recently used tracks (clock with arrow). Favorite tracks (star): you can mark your favorite tracks from the Track menu (in “Available tracks” menu, find the track and select the 3 horizontal dots next to the track name and select “Add to favorites”). Click on the clock or star icons to access recently used or favorited tracks
9) Red circular plus (+) icon in the “Available tracks” menu → add connection or track
   a) Also see top left menu in the black bar → “FILE” → Open track/connection options
   b) Also see 3-line hamburger menu at the top of the “Available tracks” menu, next to “Filter tracks” → open faceted track selector, Add track, Connections, Sort, Collapse
10) Menu options (File, Add, Tools, Help)

> TIP: Undo and redo: you can undo/redo any action via “TOOLS” → “Undo”/“Redo”, or use keyboard shortcuts Ctrl+Z/Cmd+Z (undo) and Ctrl+Y/Cmd+Shift+Z (redo)

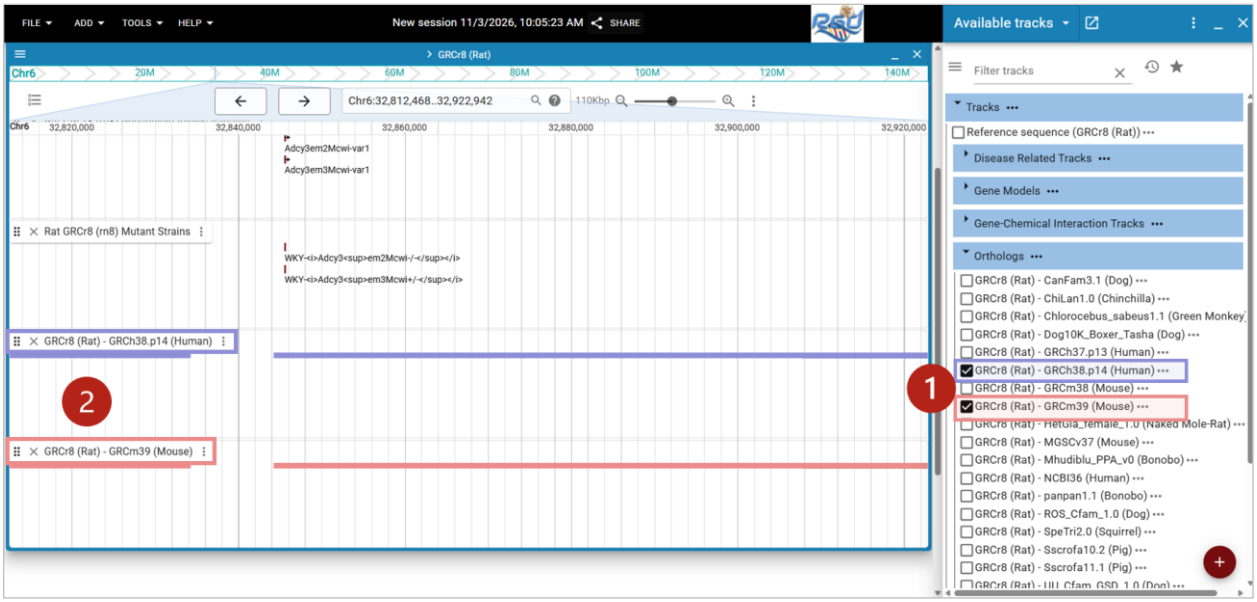
Variant Tracks



https://rgd.mcw.edu/jbrowse2/?session=share-dc_9a-x7yx&password=T9Bu3

1) **Damaging Variants**
   a) In the "Available tracks" menu, open the "Variants" → "Damaging Variants" subcategories. Select the "WKY_N (2020)_damaging" track
   b) Damaging variant in WKY/N strain in *Adcy3* (RGD:606061941)
      i) Clicking on the variant will provide "Feature details" in the right sidebar. You can also search the RGD ID number on the RGD site for more information
2) **Phenotypic Alleles Variants**
   a) In the "Available tracks" menu, open the "Variants" → "Phenotypic Alleles Variants" subcategories. Select the "Rat GRCr8 (rn8) Phenotypic Alleles Variants" track
   b) Here we see 2 variants, including Adcy3em2Mcwi-var1 associated with the Adcy3^{em2Mcwi} allele carried by the WKY-*Adcy3*$^{em2Mcwi-/-}$ strain
      i) Clicking on the variant will provide "Feature details" in the right sidebar
3) **Strains**
   a) In the "Available tracks" menu, open the "Strains" subcategory. Select the "Rat GRCr8 (rn8) Mutant Strains" track
   b) Here we see 2 mutant rat strains: WKY-*Adcy3*$^{em2Mcwi-/-}$ and WKY-*Adcy3*$^{em2Mcwi+/-}$
      i) Clicking on the strain will provide "Feature details" in the right sidebar, including phenotype terms (MP) annotated to the strain

Ortholog Tracks

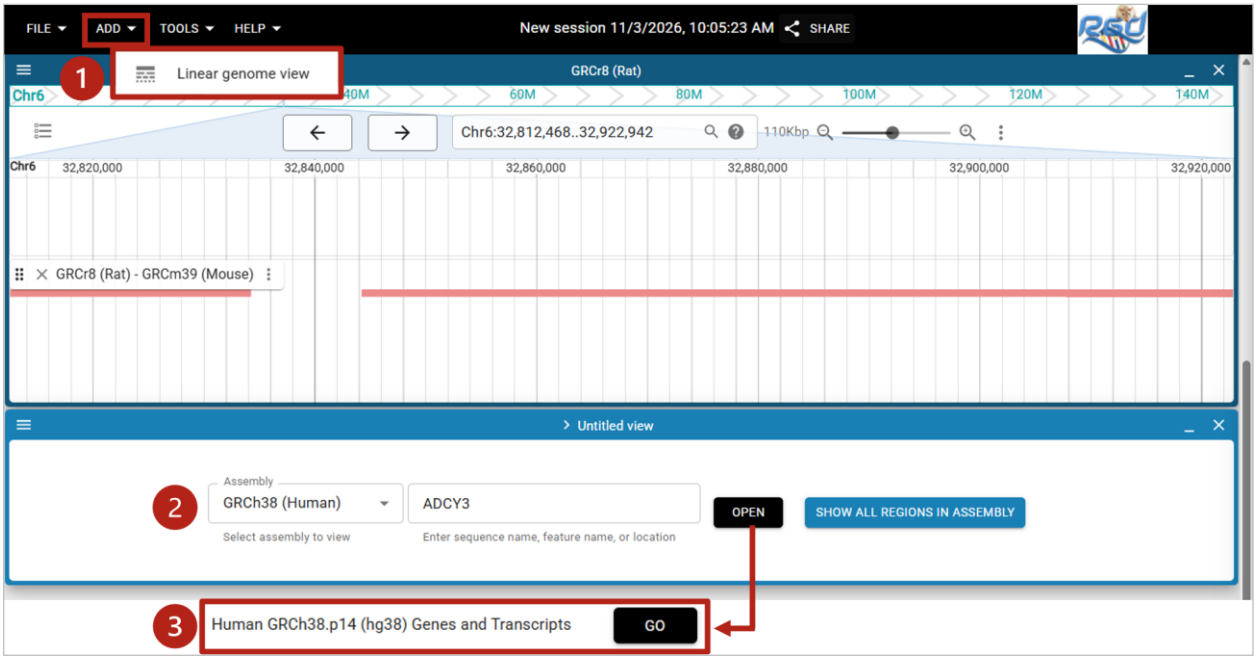


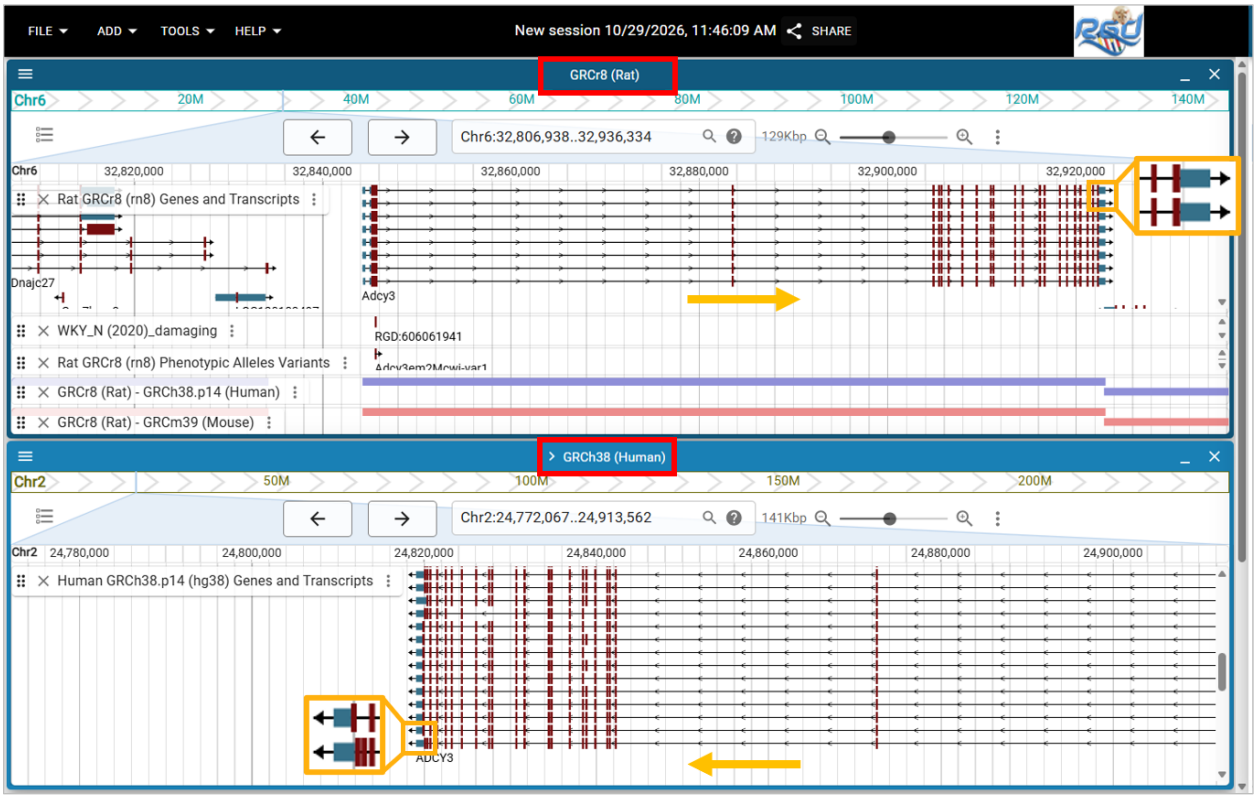
https://rgd.mcw.edu/jbrowse2/?session=share-lhsxlCUeM7&password=BQQpJ

In the “Available tracks” menu, open the “Orthologs” subcategory.

1) Select the “GRCr8 (Rat) – GRCh38.p14 (Human)” and “GRCr8 (Rat) – GRCm39 (Mouse)” tracks
2) The purple ortholog glyphs for human indicate that those genes are reversed compared to the genes in rat. The orange glyphs for mouse indicate that mouse genes are in the same orientation as rat.
    a) Click on the block glyph to bring up “Synteny feature details” in the right sidebar. This display provides the positions and lengths of the corresponding genes in the two assemblies

Linear Genome Views





https://rgd.mcw.edu/jbrowse2/?session=share-oZBXSIG2I9&password=BQJpx

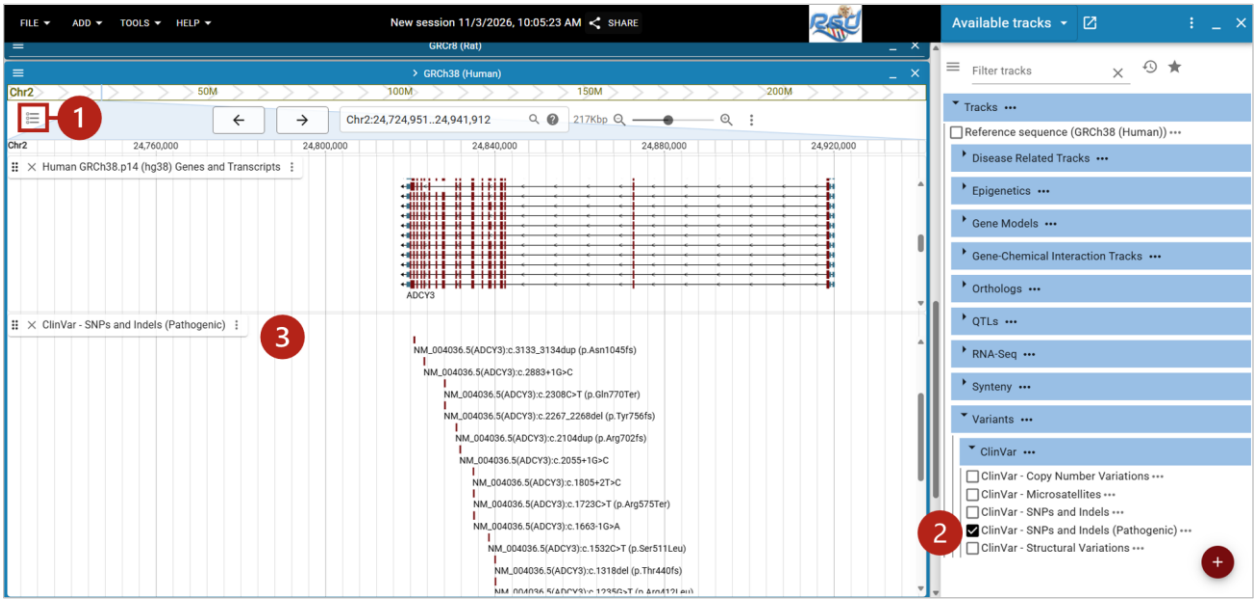
To add the human linear genome view in the same session:

1) Top left menu select “ADD” → “Linear genome view”
2) Select the assembly (e.g., human) and location (e.g., ADCY3) → click “OPEN”
3) Select “Human GRCh38.p14 (hg38) Genes and Transcripts” track from the resulting list by clicking “GO”

Now there are 2 sub-windows with the GRCr8 (Rat) assembly open at the top and the GRCh38 (Human) assembly open at the bottom. The arrows on the transcript icons indicate human genes are reversed compared to rat (yellow callout boxes and arrows).

> *Note: The gene symbol for each gene is displayed below the transcript glyphs. Hovering over a gene region will highlight the gene in gray and a popup will display the gene symbol.*

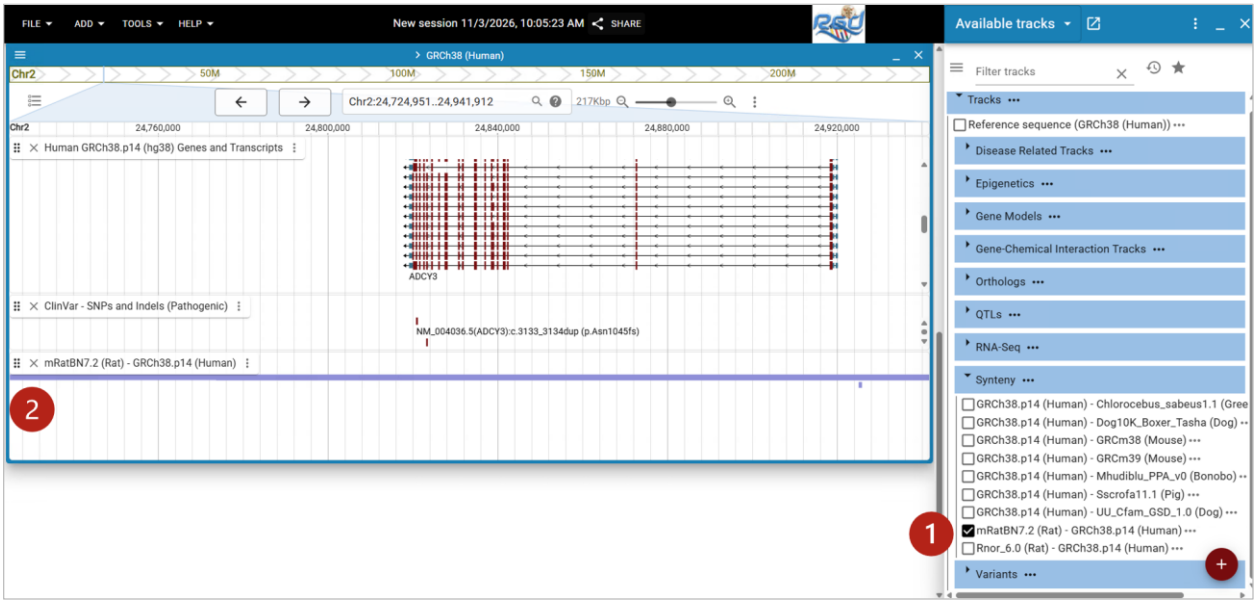
Human Variant Track



https://rgd.mcw.edu/jbrowse2/?session=share-hzRSx-8gLx&password=apjFp

1) Click the hamburger menu (3 horizontal lines beginning with a circle) in the top left of the human panel to open the “Available tracks” menu for human
2) In the “Available tracks” menu, open the “Variants” subcategory → “ClinVar” → select “ClinVar – SNPs and Indels (Pathogenic)” track
3) Scroll down (if needed) to see the *ADCY3* ClinVar variants designated as pathogenic or likely pathogenic

Rat – Human Synteny Track

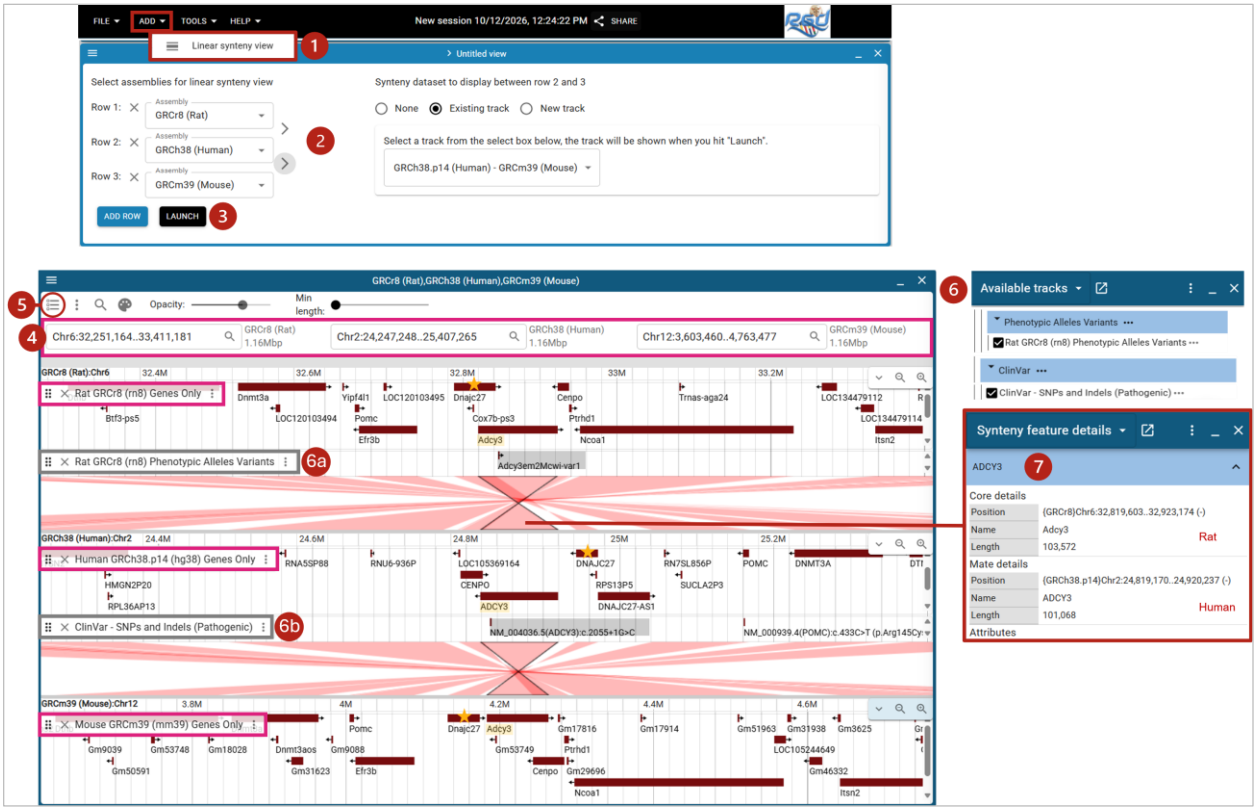


https://rgd.mcw.edu/jbrowse2/?session=share-XAxpmjJSEE&password=nB8ik

1) In the “Available tracks” menu, open the “Synteny” subcategory → select “mRatBN7.2 (Rat) – GRCh38.p14 (Human)” track
2) These blocks are generally large for closely related species, as seen here (33-35 Mb), reflecting the large conserved syntenic regions typical between rat and human

> *Note: The rat – human synteny track currently uses the previous assembly, mRatBN7.2.*

Linear Synteny View



https://rgd.mcw.edu/jbrowse2/?session=share-yfOx29AcPu&password=qgyVH

To explore synteny across multiple species:

1) Top left menu select "ADD" → "Linear synteny view"

2) Choose a species for each row (e.g., rat, human, mouse)

a) Default is 2 rows → click "ADD ROW" to select more assemblies

b) You have to click the bottom arrow (>) for the "LAUNCH" button to become activated

> *Note: On the right side, we will leave "Existing track" selected. You can also choose to add a new track – select that option and follow the prompts. Currently, it is unnecessary to select the dropdown option, as there is only one option.*

3) "LAUNCH" displays genome-wide synteny blocks connected with red lines (initial display not shown in this figure)

> *Note: Chromosomes are listed across the top of each section of the display with the width of each column determined by the relative length of the corresponding chromosome.*

4) Search "*Adcy3*" in each location box (top pink outline); select the NCBI "genes only" track for each species (tracks outlined in pink; *Adcy3* highlighted in yellow). This view shows a single bar glyph for each gene, consolidating these tracks

> *Note: The magnifying glass icon (🔍, in the same line as the (5) callout) allows you to modify the orientation of the search boxes (side-by-side, as shown, or stacked vertically).*
>
> *Zoom in/out and pan will move each section separately. "Link views" setting (3 vertical dots menu option next to the (5) callout) will zoom/pan sections together.*

5) Additional tracks can be added to any or all of the syntenic regions shown. To bring up the "Available tracks" menu in the right sidebar, select the menu icon in the top left (≡, red circle) → "Row track selectors" → select row

6) Select tracks from the "Available tracks" menu

a) Select "Rat GRCr8 (rn8) Phenotypic Alleles Variants" track

b) Select "ClinVar – SNPs and Indels (Pathogenic)" track

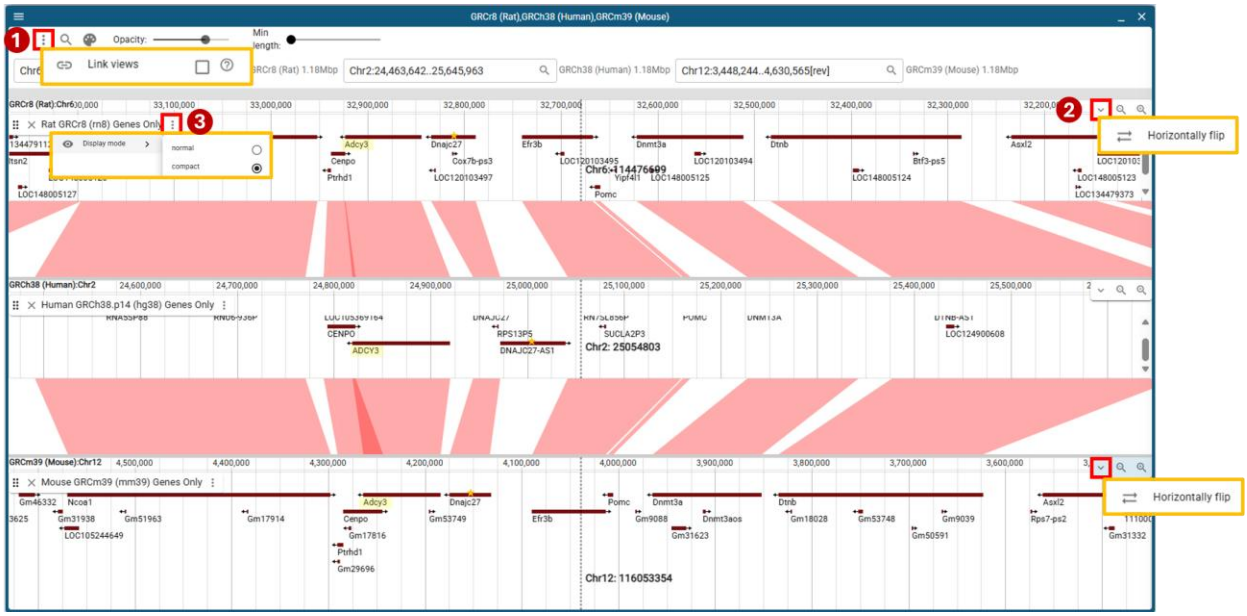
- Track names outlined in gray and variants highlighted in gray. These are the same tracks added in the previous figures

> *Note: The opacity of the red synteny tracks can be modified by the sliding bar at the top of the panel, in the same line as the (5) callout. The color of the synteny tracks can be changed by selecting the paint palette icon to the left of the opacity slider.*

7) Clicking a synteny link opens "Synteny feature details" in the right sidebar (red callout)

The human block is reversed relative to rat and mouse, as shown by gene order (*Dnajc27* starred for reference) and by the distinctive (black) X pattern of the lines in the synteny tracks that join the orthologs between the genome assembly tracks.

Bonus: uncross synteny tracks by horizontally flipping the tracks



https://rgd.mcw.edu/jbrowse2/?session=share-DMLoU9t_03&password=l5gxz

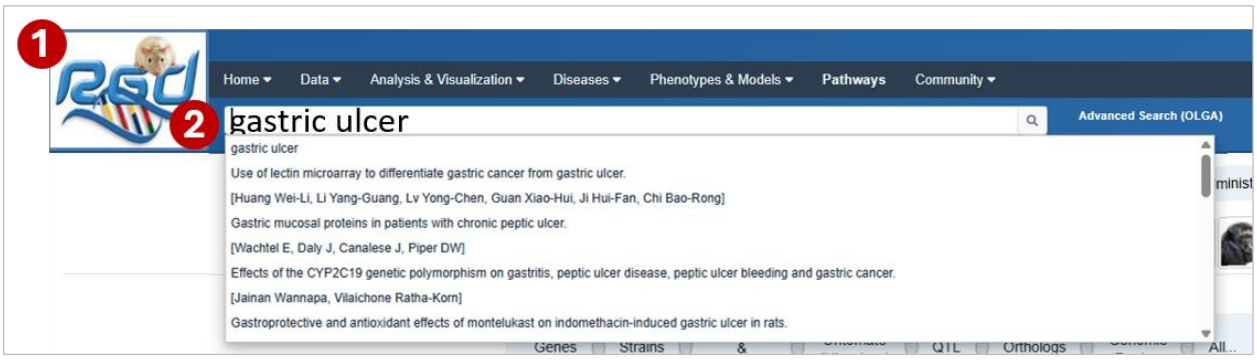
1) Uncheck "Link views" to adjust each track individually (recommended, but not required)
2) Select the down arrow next to the zoom icons → "Horizontally flip" for rat and mouse (as shown) OR just for the human track (depending on which option you choose, just note the direction of the transcript arrows and chromosome coordinates)
3) Doing this makes the gene symbols overlap. You can avoid this by selecting the 3 vertical dots menu in the track name → "Display mode" → "compact"

Adcy3 gene symbol is highlighted in yellow and *Dnajc27* is starred for reference.

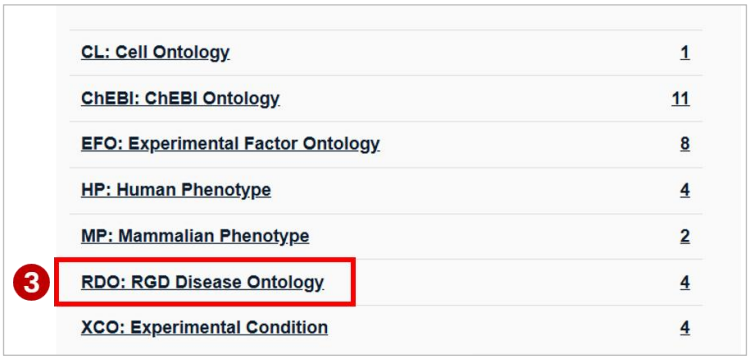
Import and view curated expression data from the RGD Downloads page

Video tutorial: https://rgd.mcw.edu/wg/viewing-expression-coverage-profiles-from-the-rgd-downloads-page/

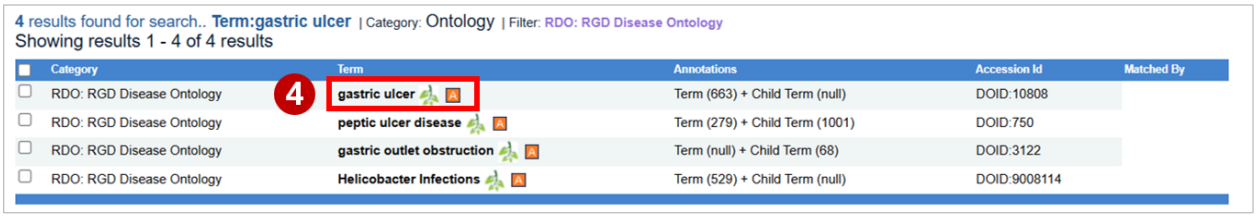
1. RGD homepage (**rgd.mcw.edu**)



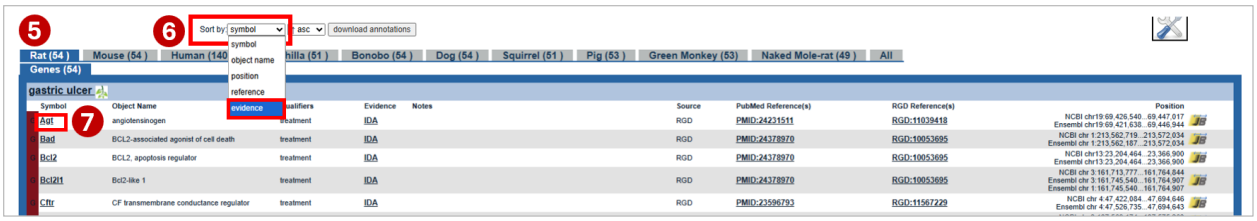
2. Search for disease of interest (e.g., gastric ulcer)
    - You could also search for a gene of interest, but for this example we will use a disease



3. Select **RDO: Disease Ontology**

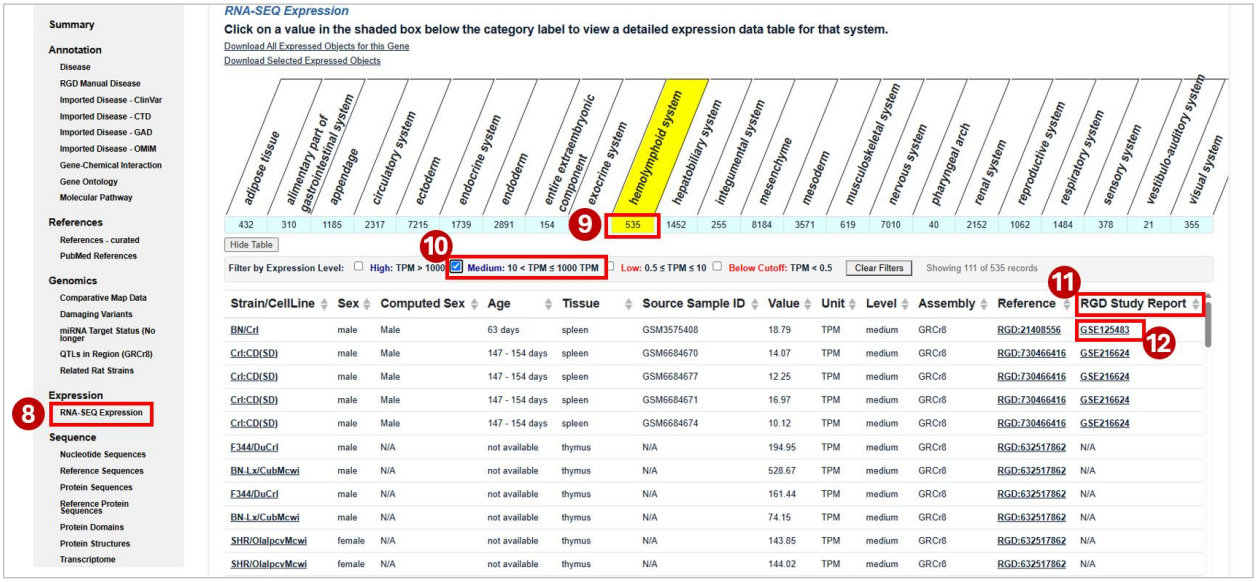


4. Select **gastric ulcer**



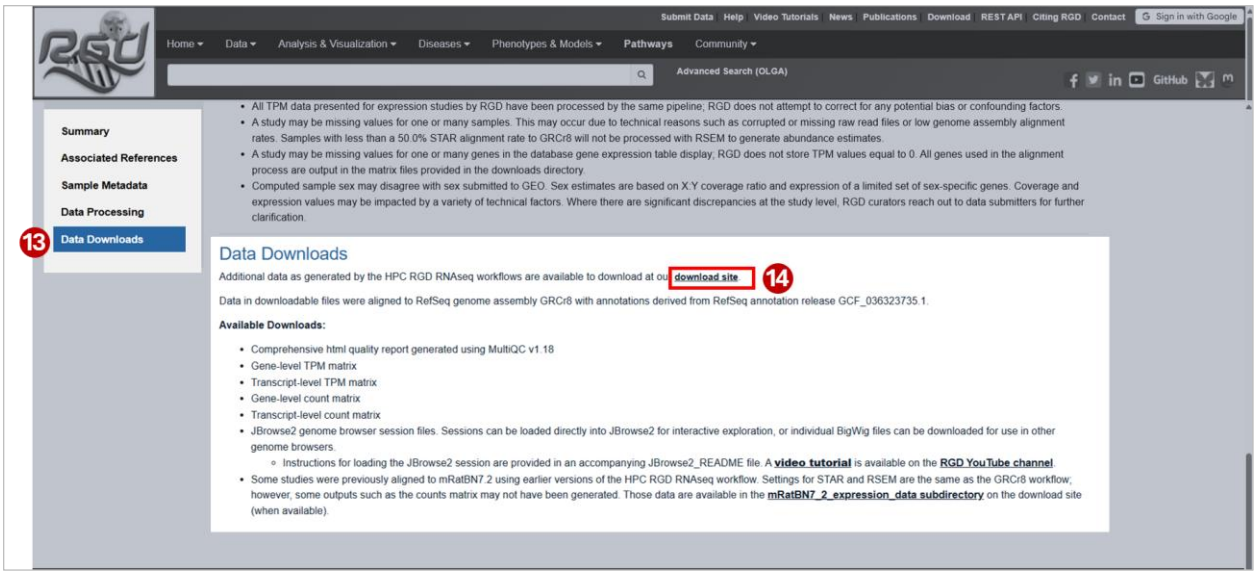
https://rgd.mcw.edu/rgdweb/ontology/annot.html?acc_id=DOID:10808&sort_by=evidence#annot

5. Scroll down on the RGD Disease Ontology annotations page for gastric ulcer
6. Sort table by **evidence**
7. Select **gene** (e.g., *Agt*)



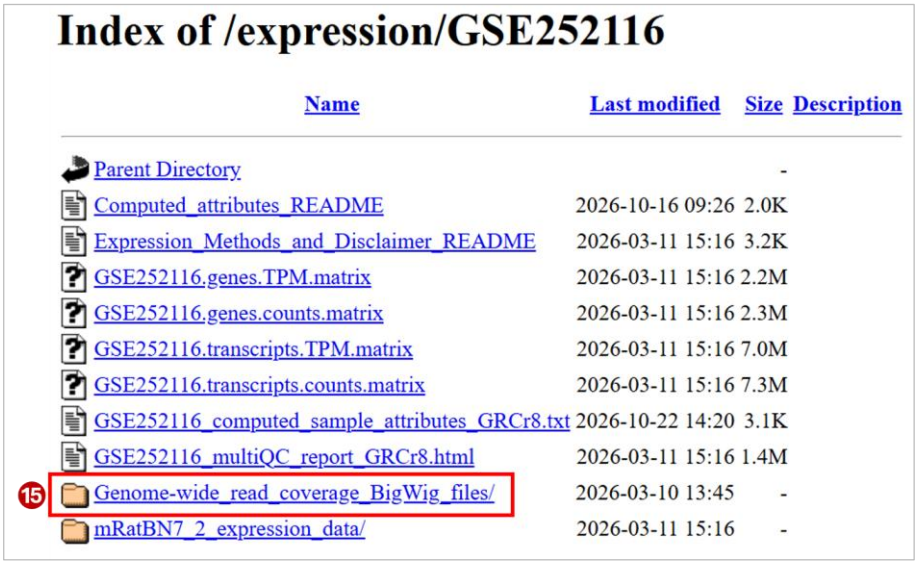
https://rgd.mcw.edu/rgdweb/report/gene/main.html?id=2069#rnaSeqExpression

8. Gene Report Page → **RNA-SEQ Expression** section
9. Select **category** (e.g., hemolymphoid system)
10. Table can be **filtered** by expression level (e.g., Medium)
11. Table can be **sorted** by column (e.g., RGD Study Report)
12. Select **GEO Study** link



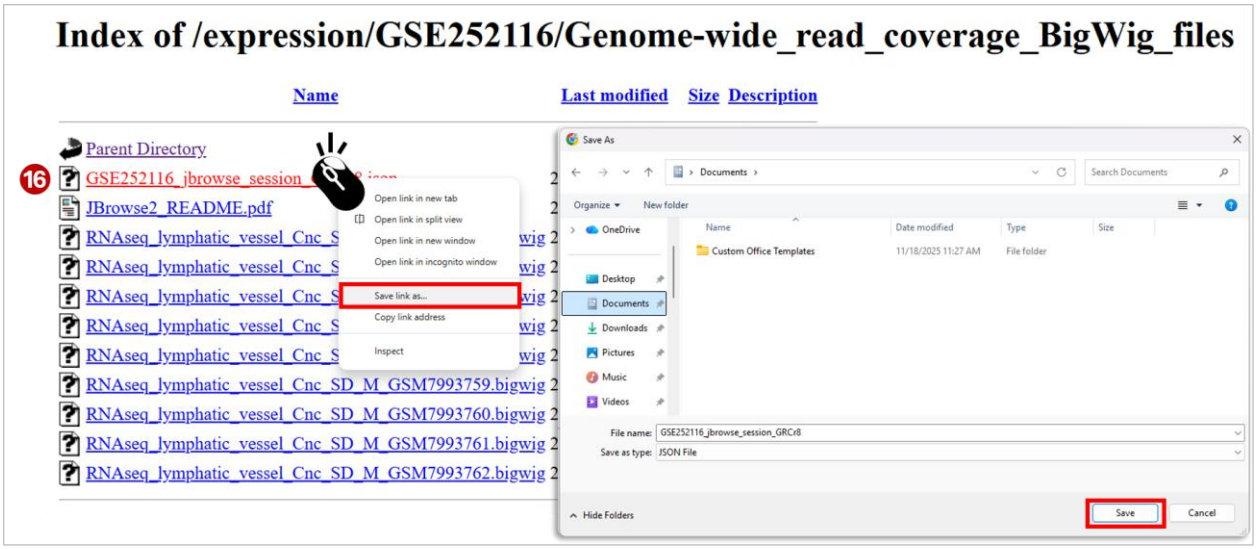
https://rgd.mcw.edu/rgdweb/report/expressionStudy/main.html?geoAcc=GSE252116#dataDownloads

13. On the Expression Study report page, navigate to **Data downloads**
14. Select **download site** link



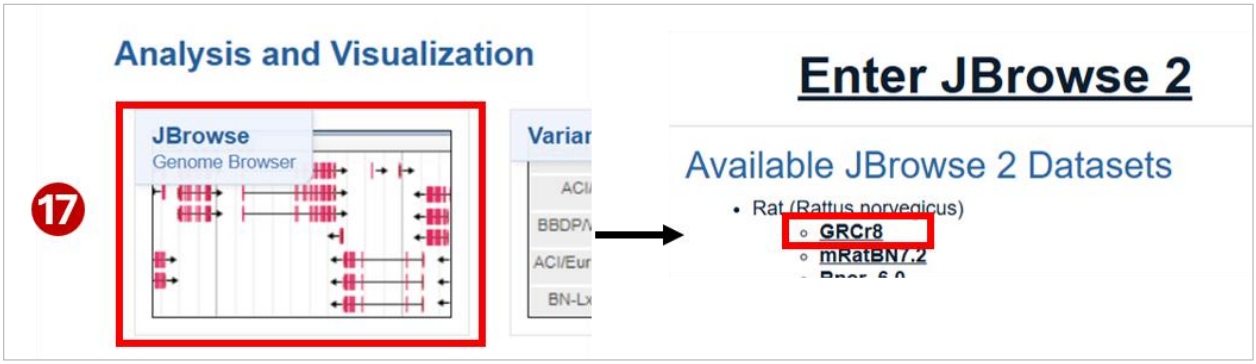
https://tinyurl.com/2b8t6jmn

15. Select **Genome-wide_read_coverage_BigWig_files/** link



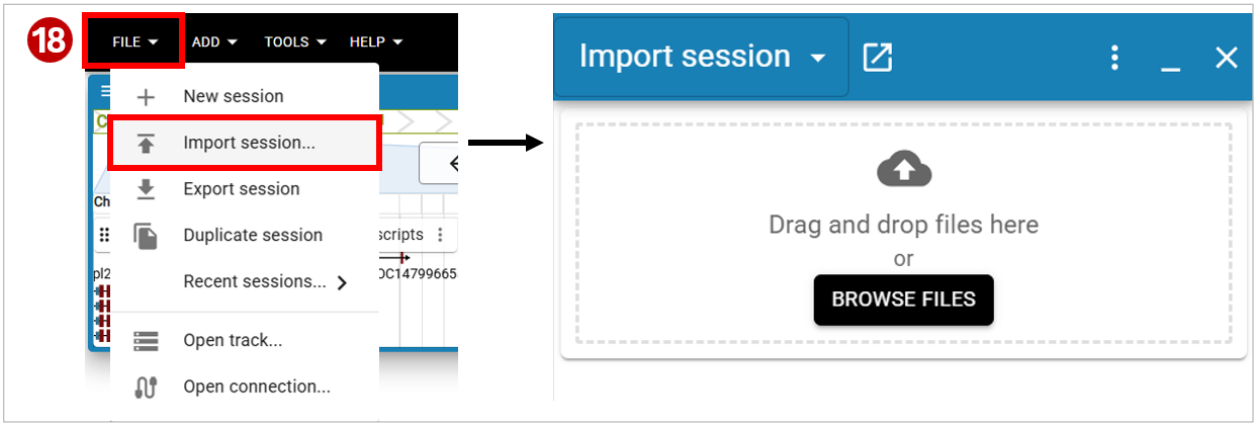
https://download.rgd.mcw.edu/expression/GSE252116/Genome-wide_read_coverage_BigWig_files/

16. To download the json file to your computer, right-click on the **GSE252116_jbrowse_session_GRCr8.json** file link and select "Save link as..." → Save

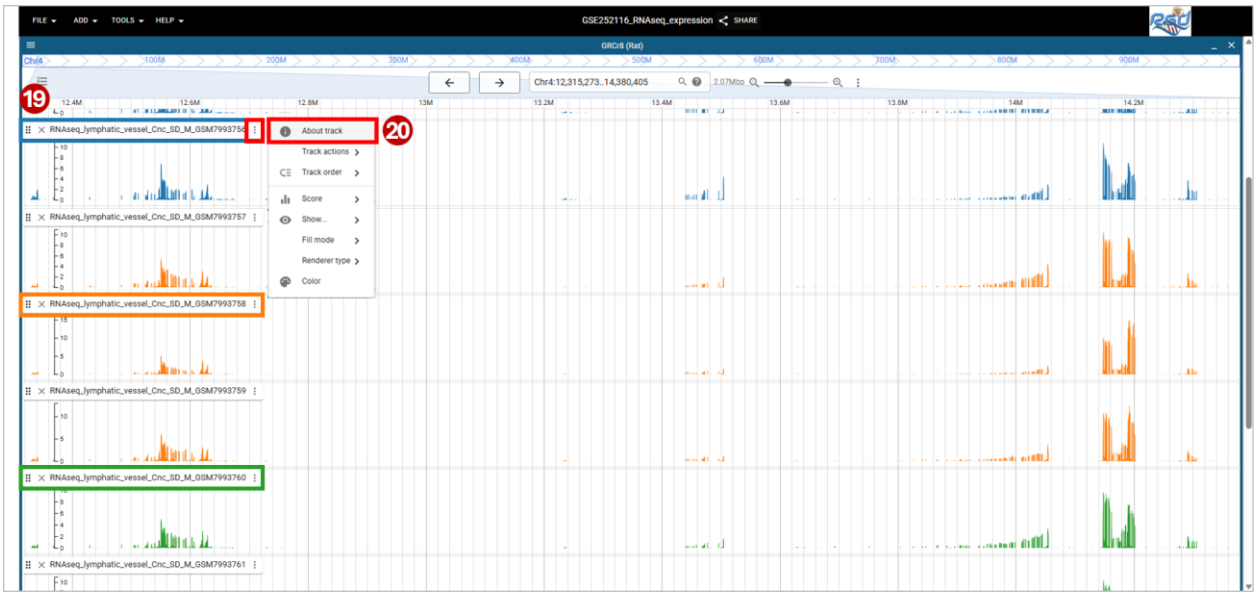


https://rgd.mcw.edu/rgdweb/jbrowse2/listing.jsp

17. Navigate to RGD **JBrowse 2** → GRCr8

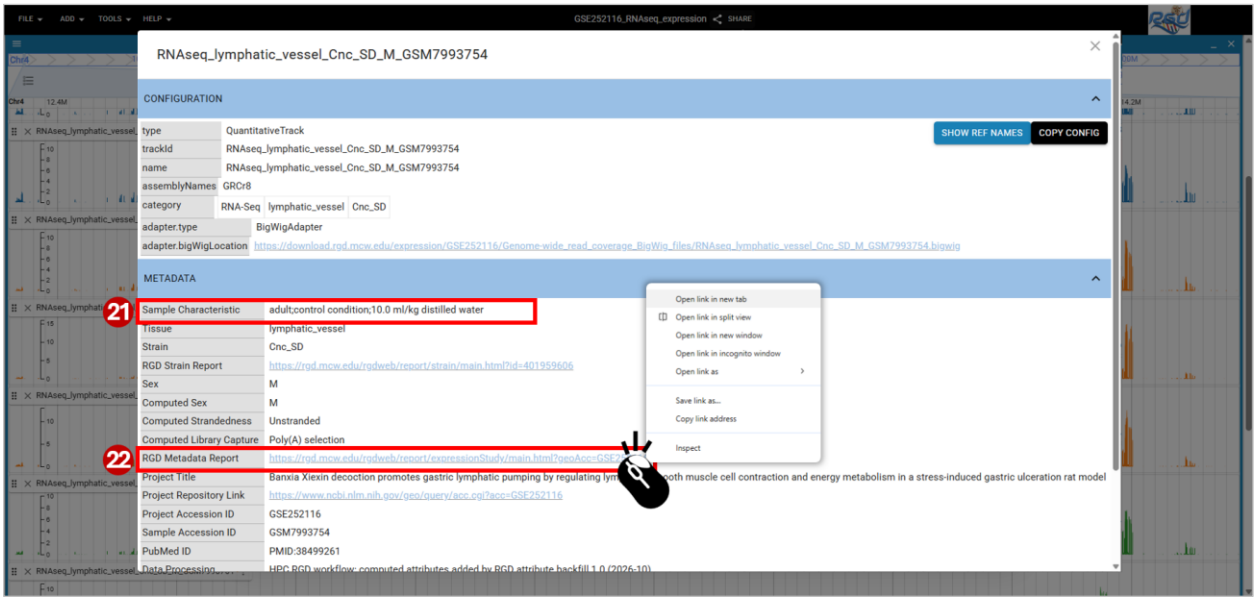


18. **Import** json file: "FILE" → "Import session" → upload json file in right sidebar

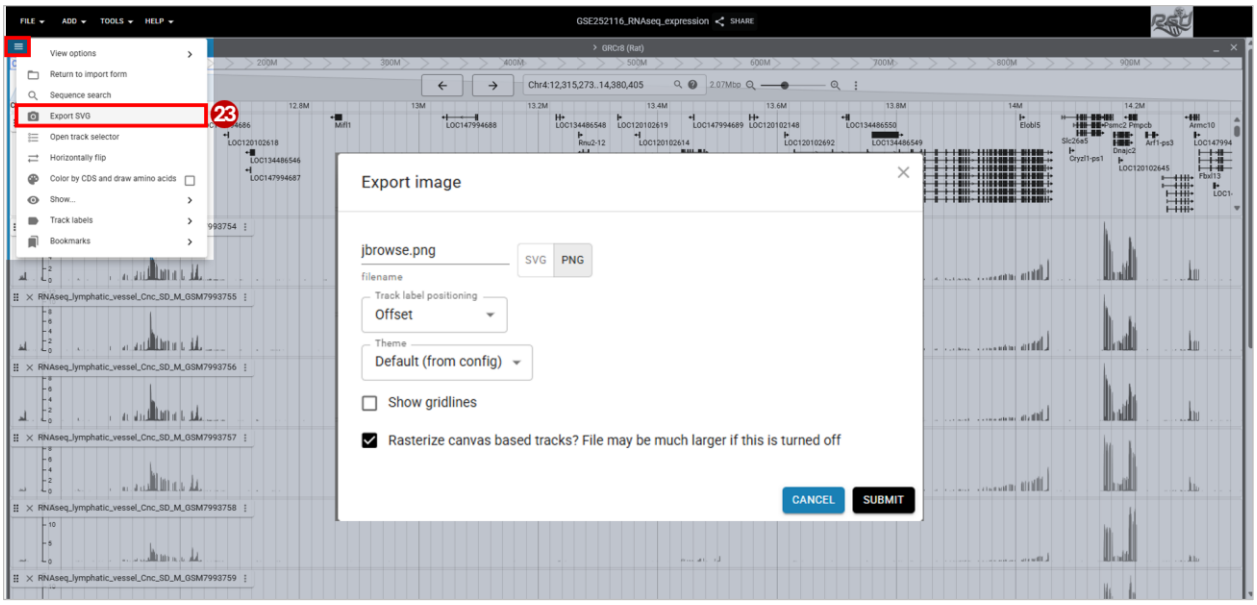


https://rgd.mcw.edu/jbrowse2/?session=share-g2fruvtads&password=zmvpw

19. **Tracks** are color-coded by matching strain, tissue, sex, and experimental conditions
    - Track names include tissue_strain_sex_GEO accession ID
20. To view experimental conditions, open the menu by clicking the 3 dots located after the track name → "About track" → Metadata display



21. Experimental conditions can be found in the "Sample Characteristic" field
22. Fully curated metadata values are available on the expression report page in the RGD user interface (linked in the "RGD Metadata Report" field)
    - Links in the metadata display will navigate away from the JBrowse 2 session, but do not open in a new tab. To avoid having to reload the session, right-click on the link and select "Open link in new tab"



23. Export JBrowse session as SVG or PNG

- At the top of the panel in the blue header, select the 3 horizontal lines menu
- Select “Export SVG”
- Select settings (e.g., switch to PNG)
- Click “SUBMIT”
- The image will download to your computer

Example PNG:

